Cantera vs NN | Test IC 32 | NRMSE [%]

T0 = 1140.7 K P0 = 3.354 atm phi0 = 1.463 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	5.24	5.76	23.7	3.74	0.329	39.4	3.17	0.324	2.22	N/A	Complete
2 P-only DeepONet10	**10k**	10.6	2.88	15.5	7.26	2.76	46.1	4.91	3.43	3.27	N/A	Complete
2 P-only DeepONet10	**long**	0.37	0.881	1.4	1.21	0.743	0.964	0.685	0.77	0.521	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	1.45	1.84	2.4	2.02	1.7	4.71	1.62	2.14	2.16	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	0.299	0.567	1.71	1.22	0.281	1.28	0.334	0.283	0.2	N/A	Complete
4 T/P DeepONet10	**10k**	6.72	5.81	15.5	7.33	5.21	47.1	3.46	5.54	28.1	N/A	Complete
4 T/P DeepONet10	**long**	77.3	19.8	156	6.73	1.25	230	17.4	2.23	66.9	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	9.03	1.91	15.4	6.68	1.13	46.5	2.91	2.45	2.15	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	7.23	1.07	10.4	1.78	0.86	41.9	3.14	0.971	0.779	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	48	4.14	50	9.62	2.13	320	25.3	7.18	15.9	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	118	22.1	211	15.8	0.877	406	31.8	78.3	1.17e+03	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	45.8	5.94	38.2	0.927	0.119	363	27.2	5.27	21.4	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	44.6	0.933	58.2	0.329	0.109	271	20.2	2.87	3.48	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.